

Mechanical Equilibrium

return to force & torque (Lectures 2 & 3)

Necessary condition for eqbm

$\Rightarrow$ resultant force & torque vanish

$$\left. \begin{aligned} \underline{r}[\Omega] &= \underline{r}_b[\Omega] + \underline{r}_s[\partial\Omega] = \int_{\Omega} \rho \underline{g} dV + \int_{\partial\Omega} \underline{t} dA = 0 \\ \underline{\tau}[\Omega] &= \underline{\tau}_b[\Omega] + \underline{\tau}_s[\partial\Omega] = \int_{\Omega} \underline{x} \times \rho \underline{b} dV + \int_{\partial\Omega} \underline{x} \times \underline{t} dA = 0 \end{aligned} \right\} \text{ for all } \Omega \subseteq B$$

If $\underline{r}[\Omega] = 0$ then $\underline{\tau}[\Omega]$ is independent of $\underline{z}$!

These conditions are intuitive but can also be derived from more general balance laws.

Integral statements $\Rightarrow$ global, whole body

Need local expressions that hold at any $\underline{z}$

Local Mechanical Equilibrium Equations

1) Force balance:

$$\underline{\underline{t}} = \underline{\underline{\sigma}} \underline{\underline{n}}$$

$$\underline{\underline{r}}[\Omega] = \int_{\partial\Omega} \underline{\underline{\sigma}} \underline{\underline{n}} dA + \int_{\Omega} \rho \underline{\underline{g}} dV = \underline{\underline{0}} \quad \underline{\underline{n}} = \text{outward normal}$$

$$\text{Tensor divergence Thm: } \oint_{\partial\Omega} \underline{\underline{\sigma}} \underline{\underline{n}} dA = \int_{\Omega} \nabla \cdot \underline{\underline{\sigma}} dV$$

$$\underline{\underline{r}}[\Omega] = \int_{\Omega} \nabla \cdot \underline{\underline{\sigma}} + \rho \underline{\underline{g}} dV = \underline{\underline{0}}$$

by the arbitrary ness of $\Omega \rightarrow$ integrand is zero
(Localization)

$$\nabla \cdot \underline{\underline{\sigma}} + \rho \underline{\underline{g}} = \underline{\underline{0}}$$

$$\sigma_{ij,j} + \rho g_i = 0$$

force balance

note: i is free index

2) Torque balance

$$\underline{\underline{\tau}}[\Omega] = \int_{\partial\Omega} \underline{\underline{x}} \times (\underline{\underline{\sigma}} \underline{\underline{n}}) dA + \int_{\Omega} \underline{\underline{x}} \times \rho \underline{\underline{b}} dV = \underline{\underline{0}}$$

substituting previous result $\rho \underline{\underline{g}} = -\nabla \cdot \underline{\underline{\sigma}}$

$$\int_{\partial\Omega} \underline{x} \times (\underline{\underline{\sigma}} \underline{n}) dA - \int_{\Omega} \underline{x} \times (\nabla \cdot \underline{\underline{\sigma}}) dV = \underline{0}$$

Introduce axial tensor $\underline{\underline{R}} \underline{n} = \underline{x} \times (\underline{\underline{\sigma}} \underline{n})$

$$R_{il} = \epsilon_{ijk} x_j \sigma_{kl}$$

so that

$$\int_{\partial\Omega} \underline{\underline{R}} \underline{n} dA - \int_{\Omega} \underline{x} \times (\nabla \cdot \underline{\underline{\sigma}}) dV = \underline{0}$$

Applying the Tensor Divergence Thm

$$\int_{\Omega} \nabla \cdot \underline{\underline{R}} - \underline{x} \times (\nabla \cdot \underline{\underline{\sigma}}) dV = 0$$

localization

$$\nabla \cdot \underline{\underline{R}} - \underline{x} \times (\nabla \cdot \underline{\underline{\sigma}}) = 0$$

in index notation: $R_{il} = \epsilon_{ijk} x_j \sigma_{kl}$

$$(\epsilon_{ijk} x_j \sigma_{kl})_{,l} - \epsilon_{ijk} x_j \sigma_{kl,l} = 0$$

using the chain rule

$$\epsilon_{ijk} x_{j,l} \sigma_{kl} + \epsilon_{ijk} x_j \sigma_{kl,l} - \epsilon_{ijk} x_j \sigma_{kl,l} = 0$$

$$\Rightarrow \epsilon_{ijk} x_{j,l} \sigma_{kl} = 0 \quad \text{with} \quad x_{j,l} = \delta_{jl}$$

$$\epsilon_{ijk} \delta_{jl} \sigma_{kl} = \underline{\epsilon_{ijk} \sigma_{kj}} = 0$$

If $\epsilon_{ijk} \sigma_{kj} = 0$ then $\epsilon_{ikj} \sigma_{jk} = 0$ flip $j \leftrightarrow k$

$$\epsilon_{ijk} \sigma_{kj} + \epsilon_{ikj} \sigma_{jk} = 0$$

$$\epsilon_{ijk} (\sigma_{kj} - \sigma_{jk}) = 0$$

choose i to be distinct from j & $k \Rightarrow \epsilon_{ijk} \neq 0$

$$\boxed{\begin{aligned} \sigma_{kj} &= \sigma_{jk} \\ \underline{\underline{\sigma}}^T &= \underline{\underline{\sigma}} \end{aligned}}$$

✓

torque balance

stress is symmetric

Mechanical Equilibrium

$$\begin{aligned} \nabla \cdot \underline{\underline{\sigma}} + \rho \underline{\underline{g}} &= \underline{\underline{0}} \\ \underline{\underline{\sigma}}^T &= \underline{\underline{\sigma}} \end{aligned}$$

$$\sigma_{ij,j} + \rho b_i = 0$$

$$\sigma_{ij} = \sigma_{ji}$$

static or eqbm equations (no time derivative)

Can we learn anything interesting?